

torch.linalg.vecdot#

<https://pytorch.org/docs/stable/generated/torch.linalg.vecdot.html>

Description:

Computes the dot product of two batches of vectors along a dimension. In symbols, this function computes $\sum_i \overline{x_i} y_i$ over the dimension dim where $\overline{x_i}$ denotes the conjugate for complex vectors, and it is the identity for real vectors. Supports input of half, bfloat16, float, double, cfloat, cdouble and integral dtypes. It also supports broadcasting. x (Tensor) – first batch of vectors of shape $(*, n)$.

Function Signature

`torch.linalg.vecdot(x, y, *, dim=-1, out=None) → Tensor#`

Parameters

Keyword Arguments

Parameters

Keyword

`dim` (int) – Dimension along which to compute the dot product. Default: -1. `out` (Tensor, optional) – output tensor. Ignored if None. Default: None.

Code Examples

```
>>> v1 = torch.randn(3, 2)
>>> v2 = torch.randn(3, 2)
>>> linalg.vecdot(v1, v2)
tensor([ 0.3223,  0.2815, -0.1944])
>>> torch.vdot(v1[0], v2[0])
tensor(0.3223)
```

Detailed Documentation

Documentation

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Supports input of half, bfloat16, float, double, cfloat, cdouble and integral dtypes. It also supports broadcasting.

x (Tensor) – first batch of vectors of shape $(*, n)$.

y (Tensor) – second batch of vectors of shape $(*, n)$.

dim (int) – Dimension along which to compute the dot product. Default: -1.

out (Tensor, optional) – output tensor. Ignored if None. Default: None.